NARCIS: Authenticated Neural Coverless Image Signaling with Attack-Qualified Codebooks and Error Correction

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ABSTRACT

Selection-based coverless communication requires more than a stable image descriptor: the selected covers must jointly support the requested symbol sequence, survive channel distortions, and provide message-level security. This paper presents Neural Adaptive Robust Coverless Image Signaling (NARCIS), an end-to-end protocol that transmits unchanged natural images. A compact self-supervised encoder maps images to normalised descriptors. A quantile codebook provides balanced symbols, while an explicit multi-attack qualification stage retains covers whose labels remain stable. Payloads and session metadata are protected with AES-GCM, symbols are assigned through a keyed locality-preserving permutation, and Reed–Solomon coding converts residual classification errors into complete-message recovery. The principal evaluation uses five deterministic partitions of 10,000 native-resolution BOSSBase images, ten encrypted messages per partition, and 21 clean, calibration, and holdout channel conditions. NARCIS recovered all 1,050 message-condition trials. Mean symbol accuracy was 98.74%, while the worst individual accuracy was 81.61% under an unseen crop. Four partitions supported 4 bits per cover and one supported 3 bits per cover. A global logistic detector and a non-linear residual-feature detector produced mean AUCs of 0.518 and 0.514, respectively; both confidence intervals included 0.5. A controlled CIFAR-100 ablation reduced complete-message success from 100% to 43.65% when attack-based qualification was removed. The principal contribution is a measurable feasible operating point for authenticated coverless communication, defined by finite-index stability, protected-message demand, and exact end-to-end recovery.

1. Introduction

Coverless image steganography represents information through natural images or generated visual content without modifying an existing carrier [1, 2]. Selection-based methods commonly associate image features with message symbols and retrieve a sequence of images from a shared database [3, 4]. Their practical performance depends on three distinct properties: the diversity of the symbol space, the stability of each selected image under transmission, and the ability of the available covers to realise the complete coded message. Reporting nominal hash length alone is insufficient to characterise these properties in practice, as high-capacity block and category mappings may still leave message-dependent coverage gaps [5, 6].

Recent work has improved one or more parts of this problem. Single-cover schemes derive many symbol locations from image blocks [5, 6]. Deep and generative systems expand the representation space [7, 8]. Stability-aware quantisation and clustering can maintain high image-level robustness at large nominal capacities [9, 10]. These advances do not remove the need for a protocol-level treatment of incomplete buckets, channel errors, payload confidentiality, metadata integrity, and replay.

This paper studies the following operational question:

> Given a finite shared image index and a declared channel model, what is the largest symbol alphabet that can carry a protected payload using unchanged covers and still recover the complete plaintext?

The proposed protocol, Neural Adaptive Robust Coverless Image Signaling (NARCIS), answers this question by selecting its operating point from measured cover stability and message feasibility. The method learns a channel-invariant descriptor, partitions its principal direction with quantile boundaries, qualifies covers against a calibration attack set, and checks the multiplicity required by the protected message before transmission. Cryptographic and coding mechanisms are then integrated into the same end-to-end path.

The contributions are:

- An end-to-end selection-based coverless protocol that combines unchanged natural covers, authenticated

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payload and metadata protection, keyed symbol assignment, replay rejection, framing, and forward-error correction.

- A feasible-operating-point rule that differs from descriptor-level robustness [9, 10]: it requires every attack-qualified bucket to satisfy the multiplicity of each protected coded message before transmission, and validates the operating point after error correction and authenticated decryption.
- A validation protocol that separates calibration attacks from holdout attack strengths and reports both symbol accuracy and complete-message success.
- A five-seed evaluation on 10,000 native-resolution BOSSBase images, covering 1,050 end-to-end trials, holdout attacks, convergence, parameter sensitivity, and linear and non-linear selection-bias tests.

Accordingly, the contribution concerns verified protected-message recovery at measured operating points under the stated datasets, attacks, detector classes, and threat model; it is not a claim of maximum nominal capacity.

2. Related Work

2.1. Hash- and retrieval-based coverless communication

Early coverless systems mapped robust image hashes to message fragments [1, 3]. Later retrieval-based schemes used local descriptors to improve resistance to common processing [4]. These methods preserve the transmitted image, but the database must contain sufficiently many stable representatives for every message symbol.

Abdulsattar [5] increased single-image nominal capacity by dividing a grayscale cover into overlapping blocks and mapping eigenvalue comparisons to ASCII values. WYSAWIS [6] instead groups 15-bit block hashes into 16 sequence categories and uses cloud-file rankings to represent segments. Both works illustrate that a large nominal representation space does not by itself guarantee that a given cover or index contains every required symbol with sufficient multiplicity.

2.2. Learned, joint, and stability-aware methods

Neural feature mapping and generation can provide semantic representations or larger message spaces [2, 7]. Ren and Wu [8] combined selection and generation modules to address mapping completeness and reconstruction. Guo et al. [9] studied robustness and diversity against passive and active analysis. Guo and Ping [10] used pseudo-Zernike moments, stability-aware quantisation, and stability-regularised clustering. They reported average robustness of 99.54%, 98.64%, and 97.19% at 10, 14, and 15 bits on Holidays, VOC, and ImageNet, respectively.

NARCIS shares the stability-aware objective but differs in scope. Its primary unit of evaluation is the recovered protected message after classification, error correction, and authenticated decryption. It also selects the codebook size from finite-bucket feasibility. The numerical A direct numerical comparison with [10] is not attempted because the datasets, attack sets, and protocol overheads differ; the cited results serve as qualitative context only.

Deep image-hiding systems based on adversarial training, encoder–decoder networks, and invertible networks have established strong payload and reconstruction results [11–16]. These systems modify or generate carrier pixels and therefore address a different design space from selection-based coverless signalling. They remain relevant to evaluation practice because they analyse robustness, detectability, and decoding jointly rather than through nominal capacity alone.

2.3. Security and evaluation

No pixel modification removes embedding artefacts, but it does not by itself establish communication security. Selection may alter the distribution of transmitted content, metadata can reveal protocol state, and an active adversary may alter or replay images. Steganalysis research has shown that weak statistical differences can be exploitable [17–22]. Formal steganographic security also distinguishes distributional secrecy from cryptographic payload protection [23]. Accordingly, NARCIS separates cryptographic security mechanisms from empirical selection detectability. The latter is treated as a measured property of a specified detector, not as an unconditional security guarantee.

3. System and Threat Model

The sender and receiver share an indexed corpus $\mathcal{I} = \{(x_i, y_i)\}_{i=1}^{N}$, a symbol-assignment key, and authenticated-encryption keys. Each x_i is a natural image and $y_i \in \{0, \dots, K-1\}$ is its codebook label. The sender transmits a sequence of unchanged images. The receiver applies the same descriptor and codebook to recover labels and then decodes the protected payload.

The channel may apply JPEG compression, additive Gaussian noise, Gaussian blur, resizing, central cropping, or small rotations. The adversary knows the algorithm and may observe, alter, reorder, or replay transmissions, but does not know the shared keys and does not compromise the local index. Traffic analysis and semantic inference from image content remain outside the cryptographic claim.

The protocol targets:

1. *cover integrity*: no embedding or pixel modification by the sender;
2. *message correctness*: exact plaintext recovery or explicit failure;
3. *confidentiality and authenticity*: authenticated encryption of payload and session metadata;
4. *freshness*: rejection of repeated or out-of-order sequence numbers;
5. *channel robustness*: correction of residual label errors within the configured Reed–Solomon budget.

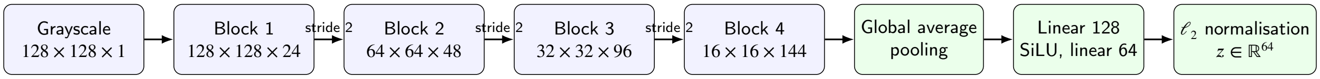

Figure 1: Descriptor encoder used in the BOSSBase experiments. Each convolutional block contains two 3×3 convolution–batch normalisation–SiLU stages. Channel transformations act on the native 512×512 image before model resizing.

4. NARCIS Method

4.1. Channel-invariant neural descriptor

Input images remain at native resolution while channel transformations are applied, and are then resized to 128×128 for descriptor inference. The encoder contains four convolutional blocks with 24, 48, 96, and 144 channels. Each block uses two 3×3 convolutions, batch normalisation, and SiLU activations; the last three blocks downsample by a factor of two. Global average pooling and a two-layer $144 \rightarrow 128 \rightarrow 64$ projection head produce a 64-dimensional unit-normalised descriptor $z = f_\theta(x)$. The network has 231,312 trainable parameters.

Training uses two independently distorted views $x^{(1)}$ and $x^{(2)}$ of the same image. Each view is generated by randomly selecting JPEG compression, Gaussian noise, blur, resize, crop, rotation, or the identity transform. The loss is

$$\mathcal{L} = \lambda_{\text{inv}}\mathcal{L}_{\text{inv}} + \lambda_{\text{var}}\mathcal{L}_{\text{var}} + \lambda_{\text{cov}}\mathcal{L}_{\text{cov}}, \tag{1}$$

where $\mathcal{L}_{\text{inv}}$ is the mean squared distance between paired descriptors, $\mathcal{L}_{\text{var}}$ prevents dimensional collapse, and $\mathcal{L}_{\text{cov}}$ penalises off-diagonal covariance. This variance–invariance–covariance structure follows the non-contrastive principle used by VICReg [24]; unlike contrastive objectives such as SimCLR [25], it does not require negative pairs. The weights are 25, 25, and 1, and the target standard deviation is $64^{-1/2}$, consistent with unit-normalised descriptors. A targeted sensitivity study in Section 5.5 compares these weights with two alternatives and dimensions 32 and 128. No semantic labels are used.

For a batch of B paired d-dimensional descriptors Z and Z', the three terms are

$$\mathcal{L}_{\text{inv}} = \frac{1}{Bd}\|Z - Z'\|_F^2, \tag{2}$$

$$\mathcal{L}_{\text{var}} = \frac{1}{d}\sum_{j=1}^{d}\left[\max(0, \gamma - \sigma_j(Z)) + \max(0, \gamma - \sigma_j(Z'))\right], \tag{3}$$

$$\mathcal{L}_{\text{cov}} = \frac{1}{d(d-1)}\sum_{i \neq j}\left(C_{ij}(Z)^2 + C_{ij}(Z')^2\right), \tag{4}$$

where $\gamma = d^{-1/2}$, σ_j is the batch standard deviation with 10^{-4} variance regularisation, and $C(Z) = (Z - \bar{Z})^\top(Z - \bar{Z})/(B-1)$. These definitions match the released implementation.

4.2. Balanced quantile codebook

Let $Z \in \mathbb{R}^{N\times 64}$ contain clean descriptors and let v_1 be the first right singular vector of the centred matrix. Each descriptor is projected as

$$s_i = (z_i - \bar{z})^\top v_1. \tag{5}$$

For a candidate alphabet of size K, boundaries are placed at the empirical quantiles j/K, $j = 1, \ldots, K-1$. The resulting label is

$$q_K(z_i) = \sum_{j=1}^{K-1}\mathbf{1}[s_i > b_j]. \tag{6}$$

Quantile partitioning balances clean occupancy before channel qualification and avoids empty clusters caused solely by an unconstrained fit. The complete clean-image assignment is defined by Eq. (6). The first singular direction is used because it is deterministic and captures the largest descriptor variance with one scalar. In the reduced sensitivity subset it explained 72.90% of the variance; the first two directions jointly explained 96.83%. Alternative directions occasionally retained more stable covers, so v_1 is treated as a reproducible design choice rather than a stability optimum.

4.3. Attack-qualified cover index

A cover is retained only when its label is unchanged under every calibration condition:

$$\mathcal{S}_K = \left\{i : q_K(f_\theta(T(x_i))) = q_K(f_\theta(x_i)),\ \forall T \in \mathcal{T}_{\text{cal}}\right\}. \tag{7}$$

The calibration set contains 12 transformations: JPEG qualities 80 and 50; Gaussian noise with standard deviations 5 and 12; Gaussian blur radii 0.8 and 1.5; resize factors 0.75 and 0.50; crop fractions 0.05 and 0.10; and rotations of 3° and 7°.

Stability filtering can introduce visible selection bias. Seven image statistics are therefore used to estimate the probability that a cover is stable: mean, standard deviation, horizontal and vertical gradient magnitude, entropy, and the 0.1 and 0.9 quantiles. Within each symbol bucket, stable covers are ordered by clipped inverse-propensity weighted random priorities. This ordering does not change labels or pixels; it reduces deterministic preference for visually distinctive stable covers.

4.4. Protected message representation

For plaintext m and sequence number n, AES-GCM [26] produces

$$c = \text{AEAD.Enc}_{k_p}(m; \text{nonce}, \texttt{NARCIS-PAYLOAD-1}\|n). \tag{8}$$

The ciphertext envelope is framed with a protocol marker, length, and CRC-32, then encoded with Reed–Solomon parity [27]. Session metadata contains the sequence number, padding, codebook size, and cover count. It is independently sealed with AES-GCM using sender and receiver identities as associated data. The receiver accepts only a sequence number greater than the highest previously accepted value for that sender.

For $K = 2^b$, the coded bit stream is divided into b-bit symbols. A keyed permutation combines a Gray ordering, a key-derived cyclic shift, and a key-derived orientation. Adjacent symbol values consequently remain adjacent in the unshifted codebook order, while the externally observed symbol-to-label assignment depends on the key.

More precisely, let $h = \text{HMAC-SHA256}(k_s, \texttt{cluster-order})$ [28], $a = \text{int}(h_{0:8}) \bmod K$, and $r = -1$ if the least significant bit of h_8 is one, otherwise $r = 1$. For position $p \in \{0, \dots, K-1\}$, define the binary-reflected Gray symbol $g(p) = p \oplus (p \gg 1)$ and

$$\pi_k(g(p)) = (a + rp) \bmod K. \tag{9}$$

The sender maps each coded symbol through π_k; the receiver constructs the exact inverse table. Equation (9) fully specifies the mapping and separates its role from payload encryption.

4.5. Feasible operating point

Let $d_k(c)$ denote the number of occurrences of label k required to transmit protected envelope c, and let a_k be the number of unused qualified covers in bucket k. A candidate codebook is feasible when

$$d_k(c) \le a_k, \qquad \forall k,\ \forall c \in \mathcal{C}_{\text{eval}}. \tag{10}$$

Candidate sizes are evaluated from the largest to the smallest. The selected operating point is the largest K satisfying Eq. (10). This criterion distinguishes nominal symbol capacity $\log_2 K$ from the net payload rate after encryption, framing, and error correction.

4.6. Computational complexity

Let N be the index size, $A = |\mathcal{T}_{\text{cal}}|$, d the descriptor dimension, K the alphabet size, L the number of transmitted covers, and F_θ the cost of one encoder inference. Clean embedding costs $O(NF_\theta)$. Multi-attack qualification is the dominant offline term, $O(ANF_\theta + ANKd)$ when label assignment is written explicitly; the scalar quantile implementation reduces assignment to $O(AN \log K)$ after projection. PCA fitting costs $O(Nd^2)$ for $N \ge d$, and sorting the projected values costs $O(N \log N)$. Online cover retrieval and keyed mapping are $O(L)$ with pre-built buckets, while decoding adds the Reed–Solomon block cost. Storage is $O(Nd + N)$ for descriptors and indexed identifiers. Offline qualification is therefore intentionally amortised across subsequent transmissions.

Figure 2 separates offline index construction from online protected transmission and recovery.

Algorithm 1 summarises transmission and recovery.

Algorithm 1: NARCIS protected coverless transmission

Input: Plaintext m, sequence n, qualified index $\mathcal{I}$, keys
Output: Recovered plaintext or failure
Encrypt m with sequence-bound AES-GCM;
Frame the ciphertext and apply Reed–Solomon coding;
Select the largest feasible codebook K;
Map coded symbols to labels with the keyed permutation;
Select one unused qualified cover for each required label;
Transmit the unchanged cover sequence and sealed metadata;
Verify metadata and reject replayed sequence numbers;
Classify received covers and invert the keyed mapping;
Apply Reed–Solomon decoding and verify CRC;
Authenticate and decrypt the payload;
return plaintext if every verification succeeds; otherwise failure;

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5. Experimental Protocol

5.1. Dataset and partitions

The principal evaluation uses BOSSBase 1.01 [29], containing 10,000 grayscale photographs at 512×512 pixels. The dataset is kept at native resolution for JPEG, noise, blur, resize, crop, and rotation; the resulting image is resized to 128×128 only at the encoder boundary. For each seed in $\{11, 29, 47, 71, 101\}$, a deterministic permutation provides 2,000 training images and a disjoint 8,000-image cover index.

The encoder is trained for five epochs with AdamW, batch size 64, learning rate 10^{-3}, and weight decay 10^{-5}. Ten independently generated eight-byte plaintexts are evaluated per seed. Reed–Solomon coding uses 128 parity bytes and symbol repetition one. Repetition one avoids multiplying the 348–464-cover sequences; robustness is instead assigned to byte-level Reed–Solomon correction. For longer payloads, fixed framing and parity overheads would be amortised, but that regime is not inferred here. This yields 50 protected messages and 1,050 message-condition trials.

CIFAR-100 [30] is retained only for the controlled qualification ablation and the matched local-descriptor comparison reported in Section 6.6. That experiment uses three seeds, 2,000 training images, an 8,000-image index, and three messages per seed.

5.2. Channel conditions and metrics

The 12 calibration transformations are those in Eq. (7). Eight holdout strengths are used only at evaluation: Gaussian noise with standard deviations 9 and 15; blur radii 1.2 and

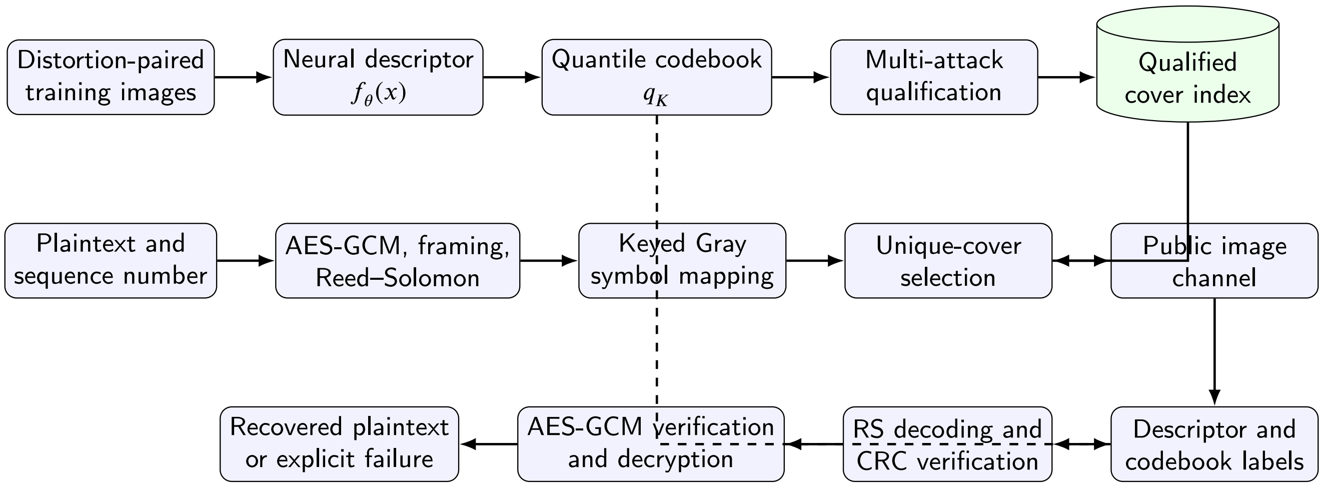

Figure 2: NARCIS construction and transmission pipeline. Training and index qualification are performed before communication. The transmitted objects are unchanged images selected from the qualified index.

1.8; crop fractions 0.08 and 0.12; and rotations of 5° and 9°. Together with the clean channel, this gives 21 conditions.

Two metrics are reported:

1. *symbol accuracy*, the fraction of transmitted covers assigned to their intended labels after channel processing;
2. *complete-message success*, which is one only when error correction, CRC verification, authenticated decryption, and plaintext equality all succeed.

Confidence intervals are two-sided 95% Student-t intervals over the five seed-level values. They quantify variation across deterministic partitions; they do not imply coverage over datasets or untested channels.

5.3. Baselines and ablations

The local baselines use a normalised 16×16 low-frequency DCT descriptor and a normalised 64-bin grayscale histogram. They are subjected to the same quantile partition, calibration attacks, candidate codebook sizes, and protected-message feasibility rule as NARCIS. This is a controlled descriptor comparison, not a reproduction of a published complete protocol.

The main ablation removes attack-based cover qualification while retaining the same neural descriptor, codebook size, protected messages, and decoder. It is run on the smaller CIFAR-100 protocol to isolate the mechanism at lower computational cost. A second check confirms the configured repetition factor of one.

5.4. Selection detectability

A deliberately interpretable detector uses seven global image statistics: mean, standard deviation, two gradient magnitudes, entropy, and two intensity quantiles. Logistic regression distinguishes selected covers from an equal number of non-selected index images using five-fold stratified cross-validation. A second detector extracts 52 high-pass residual co-occurrence summaries inspired by rich-model steganalysis [17] and applies an ExtraTrees ensemble. The area under the ROC curve (AUC) measures detectability. These tests probe qualification-induced selection bias and are stronger than a single linear detector, but they do not replace dedicated CNN steganalysis such as XuNet or SRNet [19, 22].

5.5. Sensitivity and computational measurements

A targeted reduced experiment uses seed $11, 500$ training images, a 1,500-image index, three epochs, and 96×96 model inputs. It compares descriptor dimensions 32, 64, and 128; loss weights $(25, 25, 1)$, $(1, 1, 1)$, and $(25, 10, 1)$; and projections on the first three principal directions and a fixed random direction. This experiment diagnoses design choices and is not used to replace the five-seed operating point.

Wall-clock measurements include native-image descriptor extraction, calibration, and complete transmission and recovery. Parameter counts are reported directly from the implemented models.

6. Results

6.1. Codebook feasibility

Table 1 reports the selected BOSSBase configurations. Four partitions supported 16 symbols, corresponding to 4 bits per cover. Seed 47 supported eight symbols, or 3 bits per cover. Between 30.38% and 48.73% of the index remained stable under all 12 calibration attacks.

Each protected eight-byte plaintext required 348 covers at 4 bits per cover and 464 covers at 3 bits per cover. The measured net plaintext rates were therefore 0.184 and 0.138 bits per transmitted cover. These values include encryption, framing, and error correction and must not be confused with the nominal alphabet.

6.2. End-to-end robustness

All 1,050 BOSSBase trials succeeded: five seeds, ten messages, and 21 conditions. Mean symbol accuracy was

Table 1
Selected BOSSBase operating points on 8,000-image indices.

Seed	K	Bits/cover	Stable covers	Min. bucket
11	16	4	2,823 (35.29%)	92
29	16	4	2,430 (30.38%)	61
47	8	3	3,898 (48.73%)	177
71	16	4	2,687 (33.59%)	80
101	16	4	2,846 (35.58%)	99

Table 2
Targeted sensitivity study on the reduced BOSSBase subset.

Configuration	Stable fraction	Min. bucket
64D, 25/25/1	0.335	9
32D, 25/25/1	0.375	19
128D, 25/25/1	0.412	20
64D, 1/1/1	0.284	5
64D, 25/10/1	0.357	19

98.74%. Figure 3 shows per-condition means. The two most difficult conditions were holdout crop 0.12 (84.37%) and holdout rotation 9° (91.75%). The worst individual symbol accuracy was 81.61%. Reed–Solomon corrected at most 61 byte positions, and authenticated decryption recovered every plaintext.

The clean channel, both JPEG settings, both resize settings, calibrated crops, both blur calibration settings, and rotations up to 3° produced 100% mean symbol accuracy. Message success is not inferred from these averages; it is verified after Reed–Solomon decoding, CRC checking, and authenticated decryption.

6.3. Training convergence

Figure 4 reports the complete five-epoch histories. Loss decreased substantially beyond epoch two for every seed. The final values ranged from 0.066 to 0.157. Individual curves are not monotonic because each epoch samples new distortion pairs. The experiment supports the use of five epochs over the earlier two-epoch setting, without asserting asymptotic convergence.

6.4. Selection detectability

The global detector produced mean AUC 0.518 with a 95% interval [0.489, 0.547]. The non-linear residual detector produced mean AUC 0.514 with interval [0.486, 0.543]. Figure 5 reports seed-level values. Both intervals include 0.5, and neither detector provided consistent evidence of selection above chance. These results are detector- and dataset-specific.

6.5. Sensitivity and complexity

Table 2 summarises the reduced sensitivity experiment. Equal loss weights retained fewer stable covers and produced the smallest minimum bucket. Dimension 128 improved retained stability in this subset, but a confirmatory two-seed full-protocol run recovered 419/420 trials, compared with 420/420 for the corresponding 64-dimensional runs. The 64-dimensional configuration is therefore retained because the larger representation did not improve tested capacity or reliability. The one-trial difference is not interpreted as statistically significant; it only shows that the larger descriptor provided no observed reliability gain in this limited confirmatory run.

PC1 explained 72.90% of descriptor variance and retained 33.47% of covers. PC2 retained 34.93%, PC3 31.40%, and a fixed random direction 41.27%. This result rules out a claim that PC1 is stability-optimal. It instead motivates attack-aware direction selection as a subsequent extension.

Native-image descriptor extraction averaged 87.83 seconds for 8,000 covers, or 99.85 images/s. Calibration averaged 1,246.10 seconds per partition. Complete evaluation averaged 754.63 seconds for 210 message-condition trials, equivalent to 3.59 seconds per trial. Dimensions 32, 64, and 128 contain 218,928, 231,312, and 268,368 parameters, respectively; similar inference times confirm that the convolutional backbone dominates cost.

6.6. Secondary ablation and descriptor comparison

The controlled CIFAR-100 experiment recovered 189/189 protected message-condition trials at $K = 16$, with mean symbol accuracy 98.47%. Removing attack qualification reduced complete-message success from 100% to 43.65% over the matched ablation trials. This identifies qualification as a necessary robustness component rather than treating invariant training alone as sufficient.

Under the same quantile partition, attack set, candidate sizes, and protected-message demand, neither the 16×16 DCT descriptor nor the 64-bin histogram formed a feasible codebook in any of three partitions. This is a controlled descriptor comparison, not a claim against all handcrafted coverless systems.

6.7. Position relative to published methods

Table 3 separates directly measured properties from published context. Abdulsattar and WYSAWIS provide high nominal intra-image or category-based representations [5, 6], whereas NARCIS evaluates a protected multi-cover sequence. Guo and Ping [10] report higher nominal capacities and high average robustness on larger datasets. Their reported capacity is therefore stronger than the 3–4 bit operating points measured here.

NARCIS's advantage is therefore not global capacity. Its supported positioning is protocol completeness: unchanged-cover signalling, explicit finite-index feasibility, attack-qualified selection, authenticated communication, replay control, and complete-message recovery under calibrated and holdout distortions.

Table 3
Positioning of NARCIS. Values from different publications are not treated as a common-protocol ranking.

Property	Abdulsattar [5]	WYSAWIS [6]	Guo–Ping [10]	NARCIS
Representation	Eigenvalue comparisons in overlapping blocks	15-bit block hashes grouped into 16 categories	PZM features with stability-aware quantisation and clustering	Learned invariant descriptors with quantile codebook
Primary capacity statement	High nominal single-cover location capacity	High nominal block/category capacity	10, 14, and 15 bits on Holidays, VOC, and ImageNet	3–4 bits on 8,000-image BOSSBase indices
Explicit multi-attack cover qualification	Not evaluated here	Not evaluated here	Yes	Yes
Message-level error correction	Not part of the compared mechanism	Not part of the compared mechanism	Not stated in the reported context	Reed–Solomon, measured after complete decoding
Authenticated payload and metadata	Not part of the compared mechanism	Not part of the compared mechanism	Not stated in the reported context	AES-GCM with sequence-bound data and replay guard
Holdout attack strengths	Not evaluated here	Not evaluated here	Not established from the cited summary	Eight holdout strengths
Measured complete protected-message recovery	Not evaluated here	Not evaluated here	Not directly comparable	1,050/1,050 trials

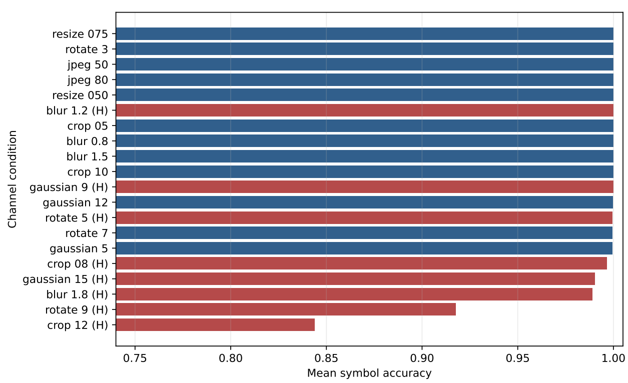

Figure 3: Mean BOSSBase symbol accuracy over five seeds and ten messages per seed. Holdout strengths were excluded from codebook qualification. Complete-message success was 100% for every condition.

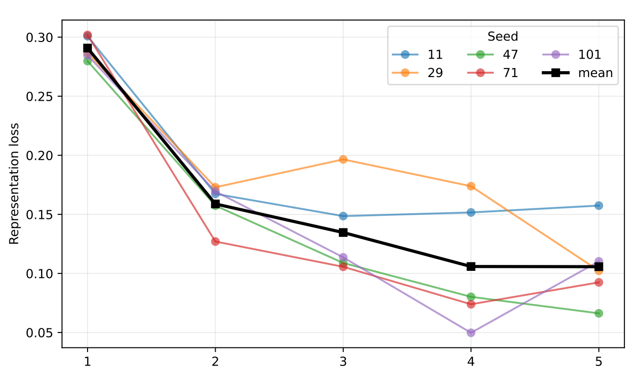

Figure 4: BOSSBase training loss for the five deterministic partitions.

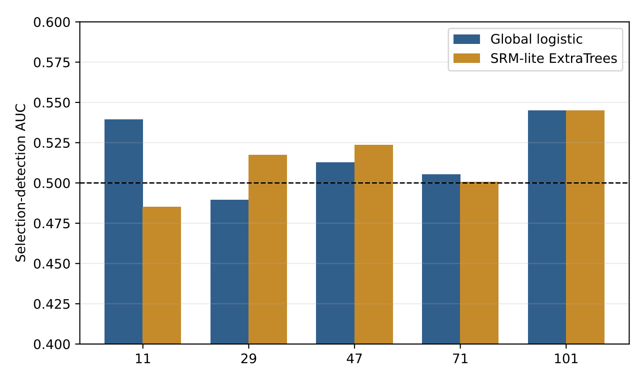

Figure 5: Selection-detection AUC by BOSSBase partition. The dashed line is chance performance.

7. Discussion

7.1. What the experiments establish

The experiments support four conclusions. First, stable descriptor averages are insufficient for communication: the multiplicity of every required symbol must be checked. Second, cover qualification is the dominant robustness component in the tested pipeline. Third, forward-error correction allows message recovery at symbol accuracies substantially below 100%. Fourth, cryptographic verification changes the evaluation criterion: a trial is successful only when the exact plaintext is authenticated and recovered.

The five-seed result also quantifies finite-index variability. The same training and selection rules yield a feasible 16-symbol alphabet on four partitions and an eight-symbol alphabet on one. Reporting this lower operating point avoids selecting only favourable partitions.

7.2. Security interpretation

AES-GCM, associated data, and monotonic sequence checking provide conventional confidentiality, authenticity, binding, and replay rejection under correct key and nonce management. These properties do not conceal communication volume or image semantics. Distributional detectability is therefore evaluated separately from cryptographic payload security. The AUC experiment addresses a different question: whether the current selection policy creates detectable shifts in seven global statistics and residual summaries. These results do not support a general undetectability claim.

7.3. Limitations

The evaluation has four main limitations.

1. BOSSBase provides native 512×512 images, but it is one grayscale photographic corpus. Cross-dataset validation remains necessary.
2. The local baselines isolate descriptor feasibility; they are not faithful reimplementations of complete published systems.
3. The selection tests include a non-linear residual detector but not an end-to-end trained SRNet. Traffic-level analysis is also outside this study.
4. PC1 is reproducible and variance-maximising, but the sensitivity study shows that attack-aware projection selection may retain more stable covers.

The measured net plaintext rates, 0.184 and 0.138 bits per transmitted cover, are lower than the nominal symbol capacities because the experiments use short payloads and 128 parity bytes. This overhead is reported explicitly. Longer-block efficiency was not measured and is not inferred.

8. Conclusion

NARCIS integrates learned image descriptors, balanced codebooks, attack-based cover qualification, keyed assignment, authenticated encryption, replay control, and Reed–Solomon correction into one selection-based coverless protocol. On five BOSSBase partitions, it recovered 1,050 of 1,050 protected message-condition trials, including eight holdout attack strengths. Four partitions supported 4 bits per cover and one supported 3 bits per cover. Mean symbol accuracy was 98.74%, and the worst individual accuracy was 81.61%. Removing qualification in the controlled ablation reduced complete-message success to 43.65%.

These results support a precise position: NARCIS provides measured robust operating points and complete protected-message recovery with unchanged natural covers. Operationally, the results show that finite-index feasibility and byte-level correction can convert imperfect label stability into exact, authenticated recovery without modifying transmitted images. The evidence does not support a global capacity or universal steganalysis claim.

Three extensions follow directly from the observed limits. First, attack-aware projection selection should be trained on calibration data while being locked before holdout evaluation. Second, cross-dataset experiments should retain native colour and resolution through channel simulation to test domain transfer. Third, selection security should be evaluated with end-to-end CNN detectors and traffic-level features. These extensions target capacity, generalisation, and detectability separately, preserving the feasible-operating-point criterion as the common evaluation unit.

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CRediT authorship contribution statement

Stéphane Gaël R. Ekodeck: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft. **Vivien Beyala Kamgang:** Conceptualization, Methodology, Writing – review and editing. **Serge Alain Ebélé:** Conceptualization, Methodology, Validation, Writing – review and editing. **Julius Marcellin Nkenlifack:** Supervision, Project administration, Writing – review and editing.

Declarations

Ethics approval. This study does not involve human participants, animals, personal data, or clinical interventions. It uses publicly available image datasets and offline computational experiments.

Competing interests. The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. WYSAWIS, cited in the related work, shares authors with this study and is identified explicitly. It is used as qualitative published context, not as an experimental baseline or as evidence for the numerical claims reported here.

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Data availability. BOSSBase 1.01 is publicly available from its original distributor at https://agents.fel.cvut.cz/boss/, and CIFAR-100 is available at https://www.cs.toronto.edu/~kriz/cifar.html, subject to their respective terms. The datasets are not redistributed in the project repository. Deterministic partition seeds and consolidated CSV/JSON results accompany the reproducibility package.

Code availability. The NARCIS implementation, attack definitions, evaluation scripts, tests, consolidated result tables, plotting code, manuscript sources, and exact random seeds are publicly available at https://github.com/EkodeckStephane/NARCIS. Trained checkpoints are not redistributed; the repository provides the commands required to retrain the models.

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